Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/11/2005
PI Name:	Meiners, Jens-christian Ph.D.		
Project Title:	Microfluidic Single-Molecule Biosensor		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	48109-1120	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	05/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Walter, Nils G. (University of Michigan)		
Grant/Contract No.:	NNA04CD01G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The ideal biosensor is extremely sensitive, specific for the target of interest, adaptable to a wide range of target analytes, and, in particular for spaceflight applications, compact, rugged, and consumes very little energy or other resources. We propose to build and characterize a prototype biosensor based on catalytic RNA molecules on a microfluidic platform that can detect single analyte molecules with high specificity. RNA aptamers can recognize a wide variety of target molecules, such as cytokines or radiation-damaged DNA nucleotides with exquisite specificity. Target recognition is reported through the cleavage of a reporter RNA molecule, which in turn causes a change in fluorescence of attached fluorophores. This change can be detected optically down to the single-molecule level. The biosensor will be constructed on a microfluidic platform that contains all provisions for sample- and reagent handling and regeneration of sensor and reporter molecules on a single, robust and easy-to handle chip. It is manufactured in a planar multilayer soft lithography process that allows the incorporation of active elements such as valves and pumps on the chip. The chip is packaged in a transparent epoxy casing to provide mechanical stability and robust interconnects. The sensor molecule		

	itself will be anchored to a hybridization spot, allowing the in-situ assembly and regeneration of the active part of the sensor. The sensor chip will be characterized with respect to sensitivity, specificity, and reliability using a theophylline analyte, paving the way for more complex analytical applications, in which a suite of different sensor molecules establishes a comprehensive analytical profile of the sample.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Two major steps towards building a demonstrator of the biosensor have been accomplished: First, a pair of detector RNA molecules has been synthezised and characterized: A sensor molecule that specifically recognizes an analyte of interest and becomes catalytically active, and a reporter molecule that changes its fluorescent properties upon cleavage by the sensor molecule have been made and characterized in single-molecule and bluk assays. Second, all required components for the microfluidic platform for the biosensor have been individually demonstrated, includign the surface attachment chemistry for the reporter RNA molecules. The next step will be the integration of the biosensor into one compact and robust chip, and an evaluation of its performance.
Bibliography Type:	Description: (Last Updated: 06/26/2007)
Abstracts for Journals and Proceedings	Korbiak, K.; Rueda, D.; Lambert, M.; Chen, H.; Walter, N.; Meiners, J.C. "Development of an RNA-Based Microfluidic Single-Molecule Biosensor Specific to Theophylline" Annual Meeting of the Biophysical Society 2005 3268-Pos , Feb-2005
Abstracts for Journals and Proceedings	Lambert, M.; Voecker, E.; Redemann, S.; Blumberg, S.; Gajraj, A.; Meiners, J.C.; Walter, N. "Monitoring Magnesium-Induced Compaction of Single RNA Molecules by Tethered Particle Microscopy" Annual Meeting of the Biophysical Society 2005 2802-Pos , Feb-2005
Abstracts for Journals and Proceedings	Chen, H.; Meiners, J.C. "Electrical properties of pressure-actuated PDMS microfluidic valves" Annual Meeting of the Biophysical Society 2005 853-Pos , Feb-2005